Untrusted input boundaries

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask which gate has actually passed and what evidence the next gate still needs.

Use visible whitespace, exact case, trusted fixture scope and no-lookup assertions without claiming universal input security.

Keep one habit: Model shape versus access → compare distinct contracts → inspect local lookup traces → assess fresh C0 values → preserve safe regression fixtures.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **untrusted input** boundaries

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from External side effects and recovery. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of external side effects and recovery

Reference example, not the only acceptable wording: Teacher explanation: External effects can succeed while their acknowledgement is lost. Recovery must inspect committed state before repeating the operation.

The mock ledger records prepared, committed or unknown per operation ID. Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure.

Preserve compensation as a separate authorised action.

Without opening the old lesson, explain one key idea from Approval gates and idempotency. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of approval gates and idempotency

Reference example, not the only acceptable wording: Teacher explanation: Approval gates bind side effects to an actor, exact version and destination.

Idempotency prevents repeated requests with the same key from repeating a committed effect. The mock submission requires approval for request version v1 and destination D. Store key -> committed result.

Reusing the same key with the same payload returns that result; the same key with different payload is rejected.

Without opening the old lesson, explain one key idea from Logs traces and privacy. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of logs traces and privacy

Reference example, not the only acceptable wording: Teacher explanation: Logs support diagnosis, but excessive recording can expose private content. Trace IDs link stages without storing complete sensitive **input**s.

For this policy log trace ID, stage, status, duration and error category; do not log passcodes, student text or teacher answers. Restrict access and declare retention.

Pictured end goal

Explain **untrusted input** boundaries using the deciding step in this fictional service.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Case exposure check: this source example appears in ai-3/31, ai-6/18. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

External side effects and recovery

Approval gates and idempotency

Logs traces and privacy

Use the three-session teaching cycle above, with extra time as needed. **Untrusted input** must be treated according to a contract before reaching code, queries or tools. Data that resembles instructions does not acquire execution authority. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **untrusted input** boundaries

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach exact data contracts without turning validation into authority

> Keep main uppercase, pictured lowercase and selected independent cases distinct.

Definitions and complete main/local-extension contract

Untrusted input must be treated according to a contract before reaching code, queries or tools. Data that resembles instructions does not acquire execution authority.

For the toy lookup accept room IDs matching one ASCII uppercase letter plus one ASCII digit, with no trimming. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters.

Main uppercase contract and local extension	Complete rule or fixture
Input type and shape	Exact Python str, whole input matching [A-Z][0-9]
Normalization	None: no trimming, case conversion or coercion
Trusted allowed IDs in the local extension	A1 and D4, supplied by the test harness
Local records	A1 has capacity 12/source ROOM-A1; B2 exists but is not in the allowed scope
Authorized but absent ID	D4 returns review after a permitted lookup

Main uppercase contract and local extension	Complete rule or fixture
Response fields	Only status, room_id, capacity and source_id; record fields are checked
Instruction-shaped input	delete_all() fails shape; no input is evaluated as code
Observation	Lookup-call IDs and an unchanged source snapshot
Scope	Fictional records and trusted-scope fixtures, not real authentication or universal injection protection

Four gate paths and checked reasoning

Validate shape, then authorize, then look up data

SUPPLIED INPUT	CHECKS AND PERMITTED LOOKUP	LOCAL RESULT
A1	Whole shape passes; trusted scope allows; exact record exists	Answer capacity12 with source ROOM-A1
delete_all()	Whole shape fails before access or lookup	invalid_input; no instruction executes
B2	Shape passes, trusted access excludes it	denied; no record lookup
D4	Shape passes and access allows, but record absent	review; do not invent a capacity

The original worked case classifies A1 and instruction-shaped text. This explicitly supplied local extension additionally tests authority, existence and response fields.

Require exact ASCII full-match and no normalization for the main uppercase contract. A1 passes shape and the explicitly added local fixture supplies permission/data; the original format result alone was not a measured record lookup. delete_all() remains data and fails shape, B2 is denied before lookup, and authorized-but-absent D4 reviews. The picture uses lowercase rather than uppercase.

The selected current C0 task is different from retained older B2 practice: classify only its stated values and explain the separate authority gate. Keep diagnostic records within privacy policy; local code success is not a universal injection-defense claim.

Untrusted input boundaries — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Inputs A1 and delete_all() are supplied to room lookup.	For the toy lookup accept room IDs matching one ASCII uppercase letter plus one ASCII digit, with no trimming. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters.	A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string. Qualification: A format check does not grant resource access; authorisation follows separately.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Inputs A1 and delete_all() are supplied to room lookup.

A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s. The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it

establishes. Retain a policy-compliant diagnostic record of the outcome so a later revision can be compared honestly.

Whole-input, type, Unicode and access probes

Input or boundary probe	Expected interpretation
A1 under the main contract	Shape passes; only separate trusted scope and record evidence permit the example answer
a1, A11 or A1 with trailing space/newline	Main whole-input shape fails; do not trim or change case
Fullwidth characters or a non-ASCII digit	Do not match the declared ASCII ranges
None or another non-string type	Fail before regex matching; no implicit conversion
delete_all()	Instruction-shaped data fails the format and is never executed
Valid B2 outside trusted scope	Deny before lookup, even if a fixture record exists
Valid authorized D4 with no record	Review after lookup, without inventing content
Lowercase-picture z0	Shape only; authority remains a separate question
All these local probes pass	No proof that every query, shell, model or deployed input boundary is protected

Separate pictured lowercase inputs and scope

Check shape before permission. A toy tool accepts exactly one ASCII lowercase letter followed by one ASCII digit, with no trimming.

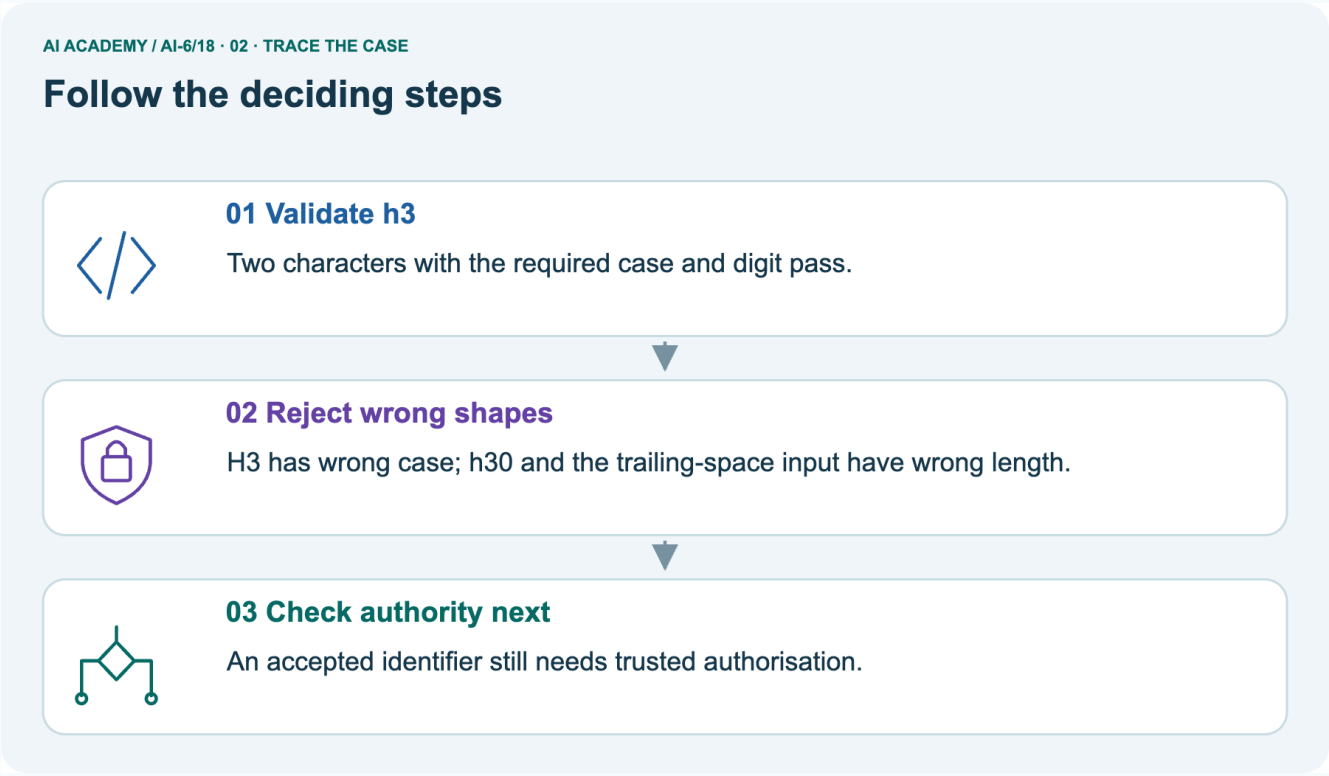
Inputs h3, H3, h30 and "h3 ". Shape acceptance never grants access by itself.

The main contract is uppercase [A-Z][0-9]. The picture is a separate lowercase [a-z][0-9] contract, also with no trimming. Under that pictured contract h3 passes; H3 has the wrong case; h30 has an extra character; "h3 " has a trailing space. The changed z0 passes shape

but still needs trusted authorization.

Do not apply the main uppercase rule to the picture or treat either shape result as permission to read a record.

Picture trace and valid-z0/no-permission key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Validate h3	Two characters with the required case and digit pass.
2. Reject wrong shapes	H3 has wrong case; h30 and the trailing-space input have wrong length.
3. Check authority next	An accepted identifier still needs trusted authorisation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Input shape and permission answer different questions. Silently normalising these **input**s would change the stated contract.

Some systems normalise **input**s explicitly; this fictional contract deliberately does not.

Change: Replace h3 with z0.

Prediction: The shape passes; resource access is still undecided.

Reason: Structure contains no **evidence** of ownership or user permission.

▼ Reference: original format-only helper

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
import re

def validate_identifier(value):
    return type(value) is str and re.fullmatch('[A-Z][0-9]', value) is not None

print(validate_identifier("B2"))
```

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Untrusted input	Data whose claims or control intent have not been authorised
Full match	Validation covering the whole input
Injection boundary	A separation preventing data from becoming executable control

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

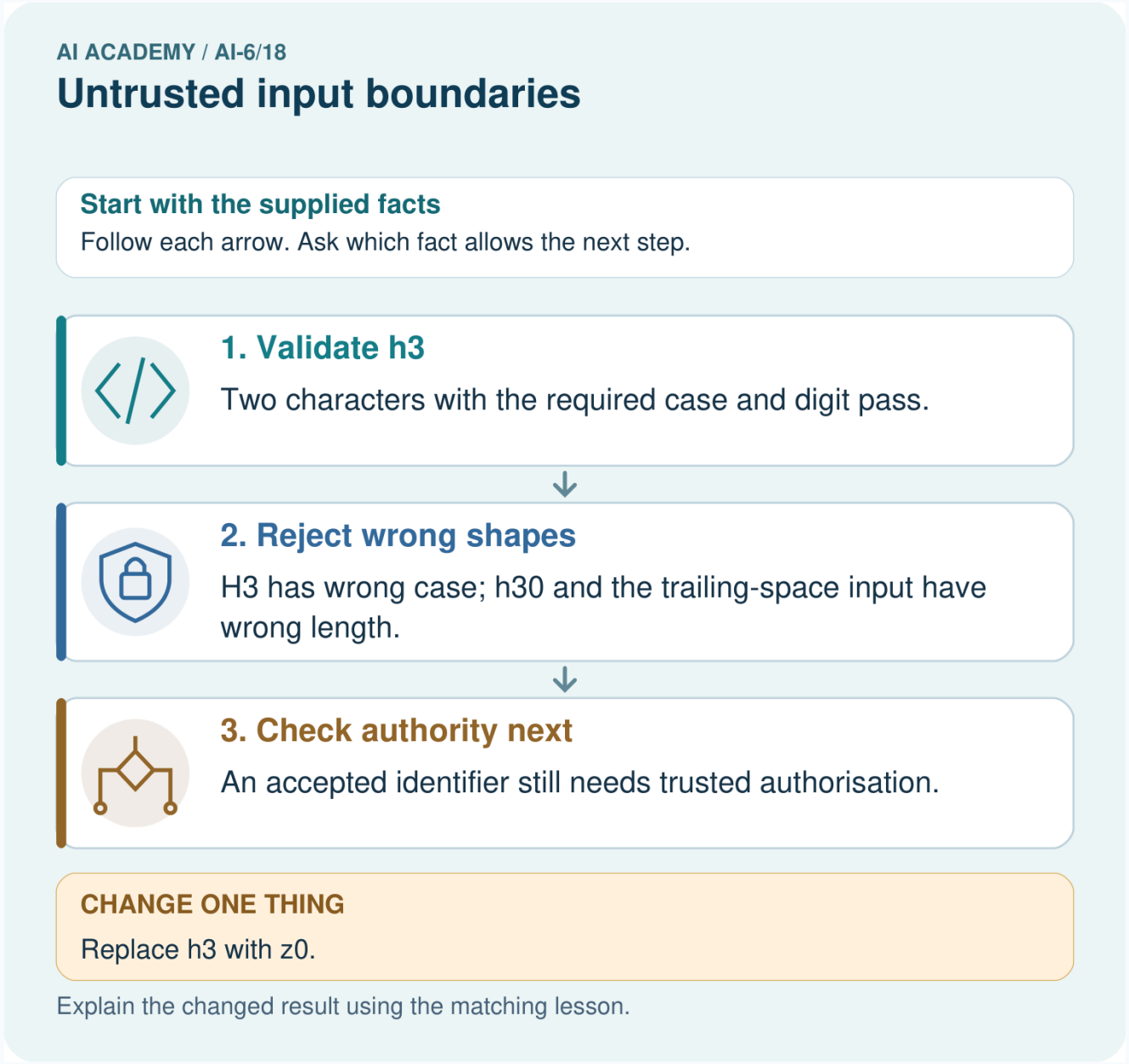
Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?

Thinking move	Teacher prompt
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Untrusted input boundaries

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: **Input**s A1 and delete_all() are supplied to room lookup. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Untrusted input** boundaries. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A format check does not grant resource access; authorisation follows separately.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Classify h3, H3, h30 and the trailing-space **input** by the stated shape contract. Does a valid z0 grant record access, and which separate gate still remains?

Two characters with the required case and digit pass. H3 has wrong case; h30 and the trailing-space **input** have wrong length. An accepted identifier still needs trusted authorisation.

Changed case: The shape passes; resource access is still undecided. Structure contains no **evidence** of ownership or user permission.

Changed-case reasoning: The shape passes; resource access is still undecided. Structure contains no **evidence** of ownership or user permission.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main A1 shape passes; exact local permission and record allow example lookup. Instruction-shaped string rejects without execution; valid B2 denies before lookup; authorized missing D4 reviews. Lowercase-picture h3/z0 shapes pass but access is separate. Selected C0 passes uppercase ASCII shape; c0 wrong case, C00 extra character, C0 with trailing space wrong whole length, None wrong type.

Valid format alone does not grant access. Use the selected C0 key, not earlier B2 practice, and do not silently trim or normalize.

Diagnose the current misconceptions

The worked result establishes every future case: A format check does not grant resource access; authorisation follows separately.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: **Input** format is exactly one ASCII uppercase letter followed by one digit, with no trimming. Evaluate C0, c0, C00, "C0 ", and None. Does a valid format grant access?

Teacher explanation: C0 passes the exact string format. c0 fails uppercase case, C00 fails length, "C0 " fails because the trailing space is not trimmed, and None fails the string-type requirement. A well-formed identifier does not grant resource access; trusted authorisation must still be checked.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: **Untrusted input** must be treated according to a contract before reaching code, queries or tools. Data that resembles instructions does not acquire execution authority. For the toy lookup accept room IDs matching one ASCII uppercase letter plus one ASCII digit, with no trimming. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters.

W2 Trace the worked case

Inputs A1 and delete_all() are supplied to room lookup. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

W3 Audit the limit

A report omits this qualification: A format check does not grant resource access; authorisation follows separately. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A format check does not grant resource access; authorisation follows separately.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A format check does not grant resource access; authorisation follows separately.

The new case should test this particular **mechanism** rather than merely restate the original result.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Guided and exit questions

Guided: A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string. Qualification: A format check does not grant resource access; authorisation follows separately.
</details>

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Classify B2, b2, B22 and B2; DROP under the exact format.

Teacher explanation: B2 valid; lowercase, extra digit and injected command text invalid. Validation and safe lookup both apply.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: B2 valid; lowercase, extra digit and injected command text invalid. Validation and safe lookup both apply. Require intermediate reasoning and the stated scope. Fresh retry: **Input**s Z9 and Z9 followed by a newline.

Expected reasoning: Z9 valid; extra newline fails exact full-match policy. Do not silently trim unless specified.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled **untrusted input** boundaries **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **untrusted input** boundaries.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Sandboxing and resource limits. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/19, ai-6/21, ai-6/24. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	

Record	Teacher evidence
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **untrusted input** boundaries

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: **Input**s A1 and delete_all() are supplied to room lookup. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

Check every learner before discussion: Use the worked case for **Untrusted input** boundaries. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A format check does not grant resource access; authorisation follows separately.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Replace h3 with z0. Require a **prediction** and a reason before comparing. Expected result: The shape passes; resource access is still undecided. Reason: Structure contains no **evidence** of ownership or user permission.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

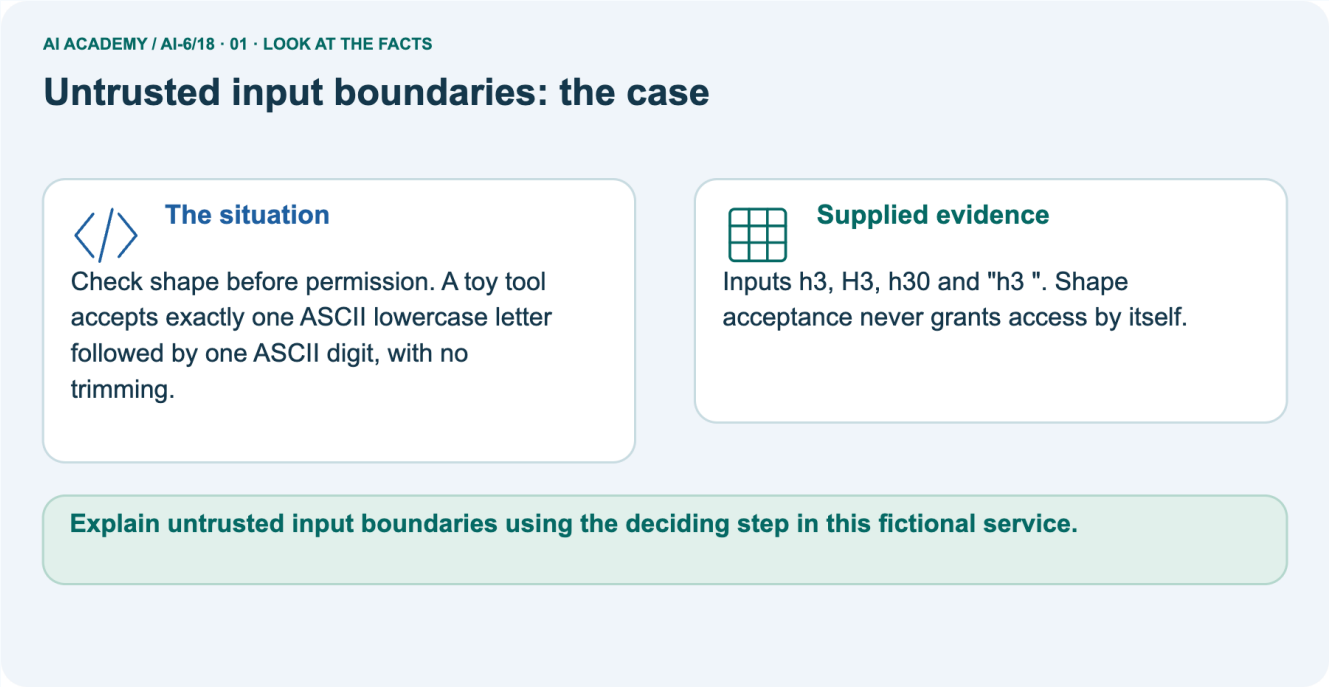
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Untrusted input boundaries: the case



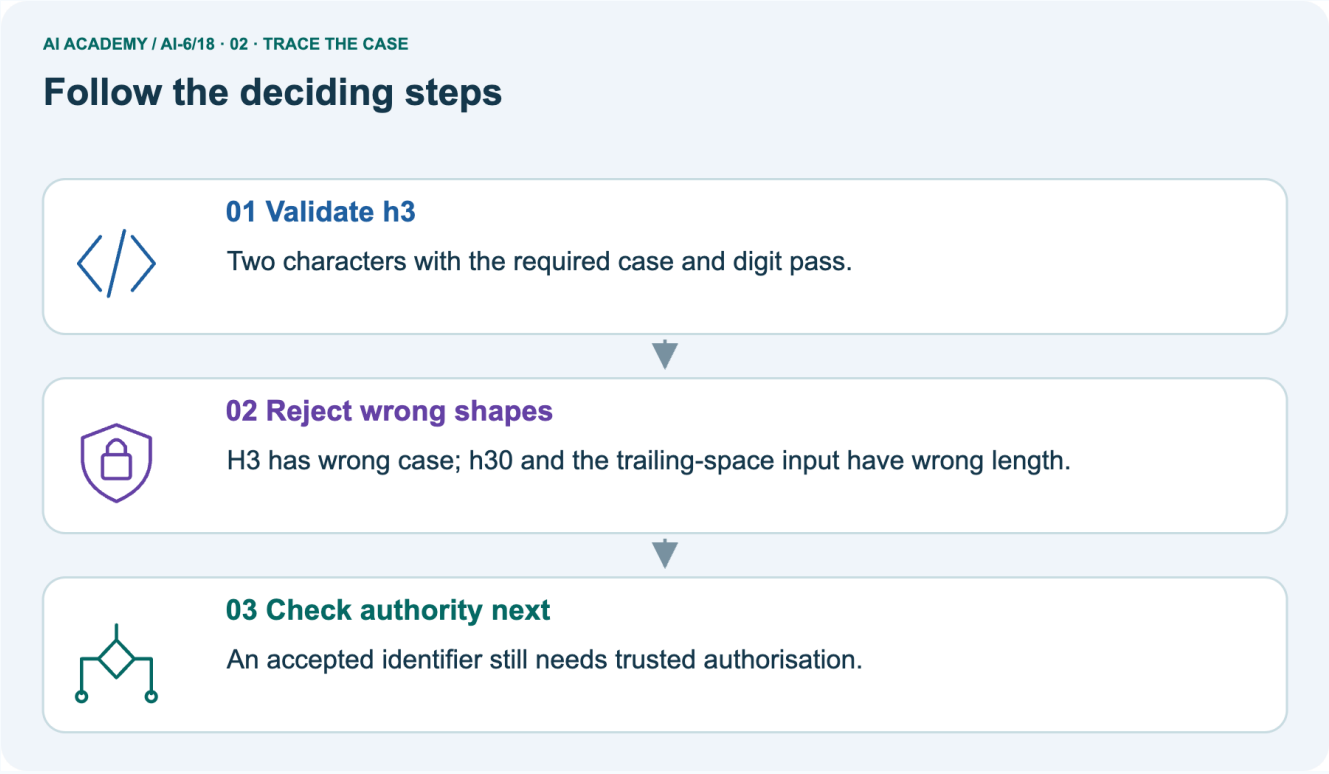
Follow the labelled picture with the matching explanation. Open the vector diagram

Check shape before permission. A toy tool accepts exactly one ASCII lowercase letter followed by one ASCII digit, with no trimming. Supplied facts: **Input**s h3, H3, h30 and "h3 ". Shape acceptance never grants access by itself.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



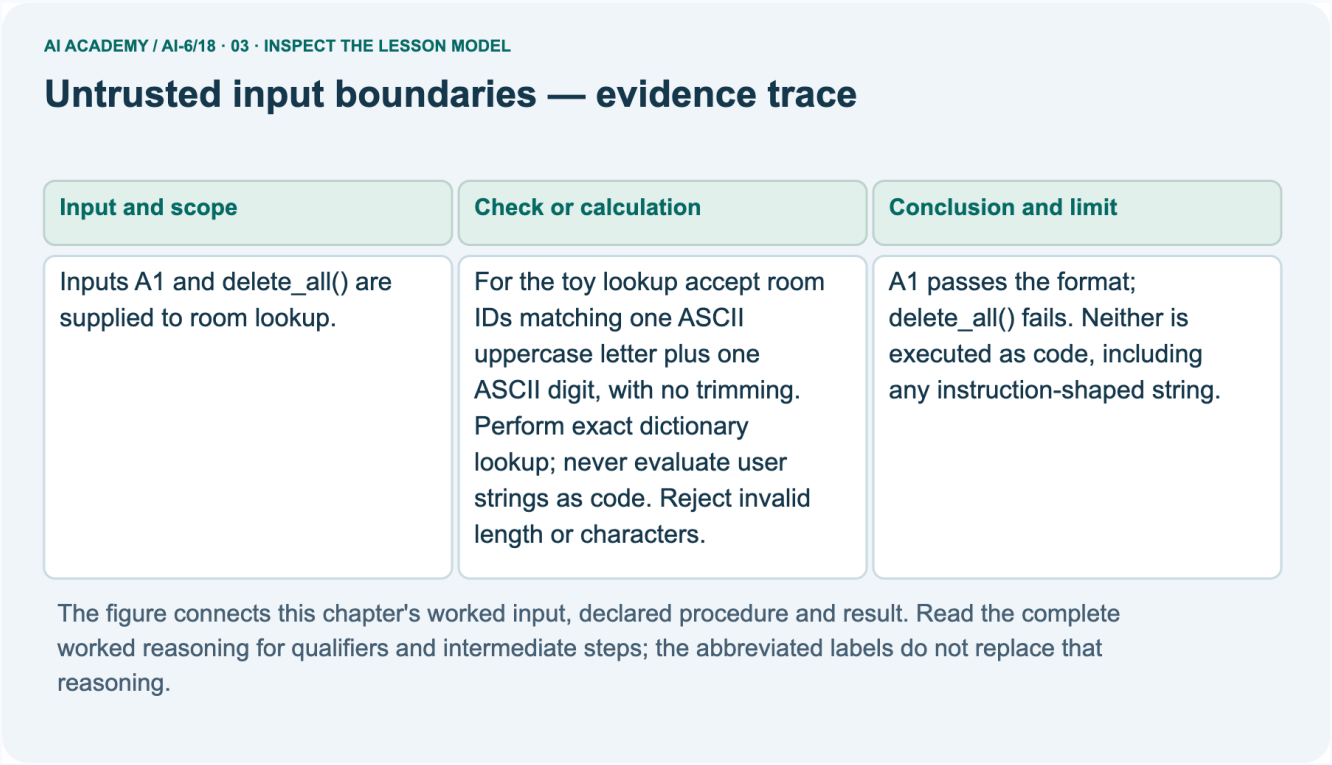
Follow the labelled picture with the matching explanation. Open the vector diagram

Validate h3: Two characters with the required case and digit pass. → Reject wrong shapes: H3 has wrong case; h30 and the trailing-space **input** have wrong length. → Check authority next: An accepted identifier still needs trusted authorisation.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Untrusted input boundaries — evidence trace



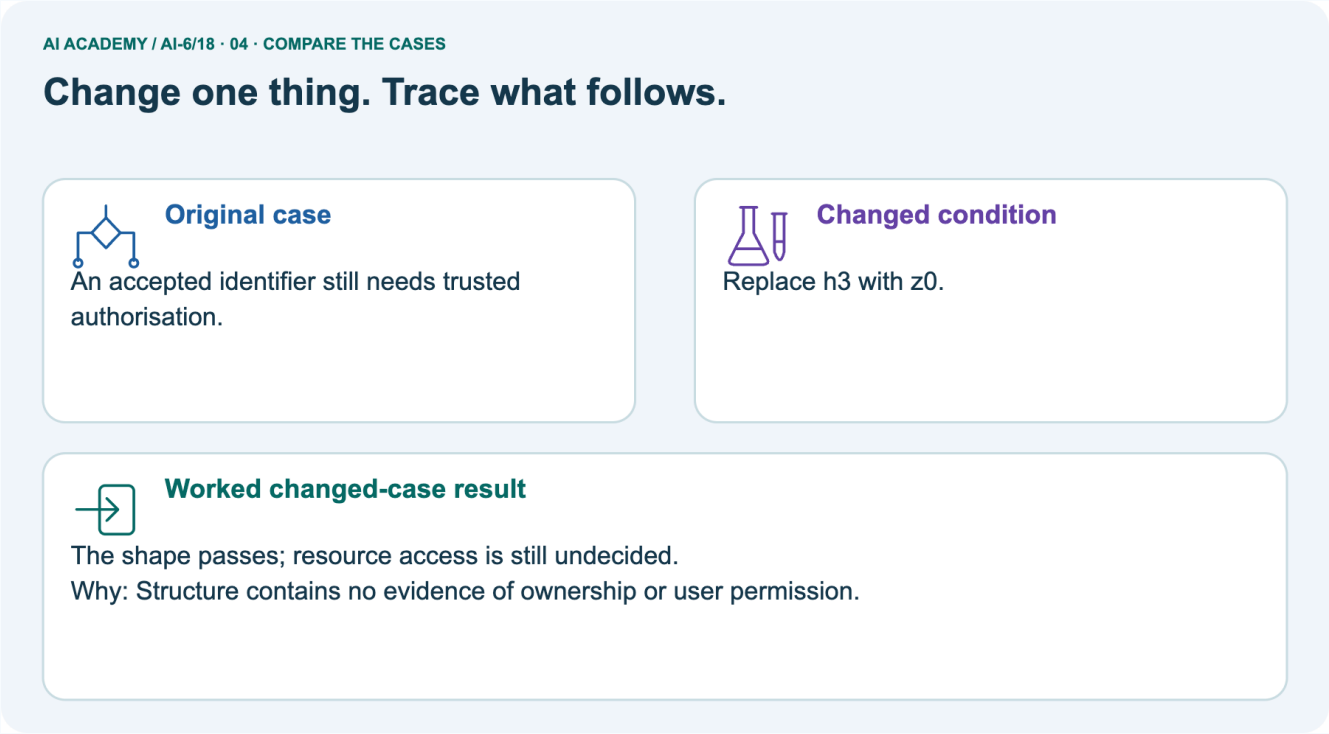
Follow the labelled picture with the matching explanation. Open the vector diagram

Untrusted input boundaries — **evidence** trace. **Input**s A1 and delete_all() are supplied to room lookup. | For the toy lookup accept room IDs matching one ASCII uppercase letter plus one ASCII digit, with no trimming. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters. | A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: An accepted identifier still needs trusted authorisation. Changed condition: Replace h3 with z0. Expected result: The shape passes; resource access is still undecided.

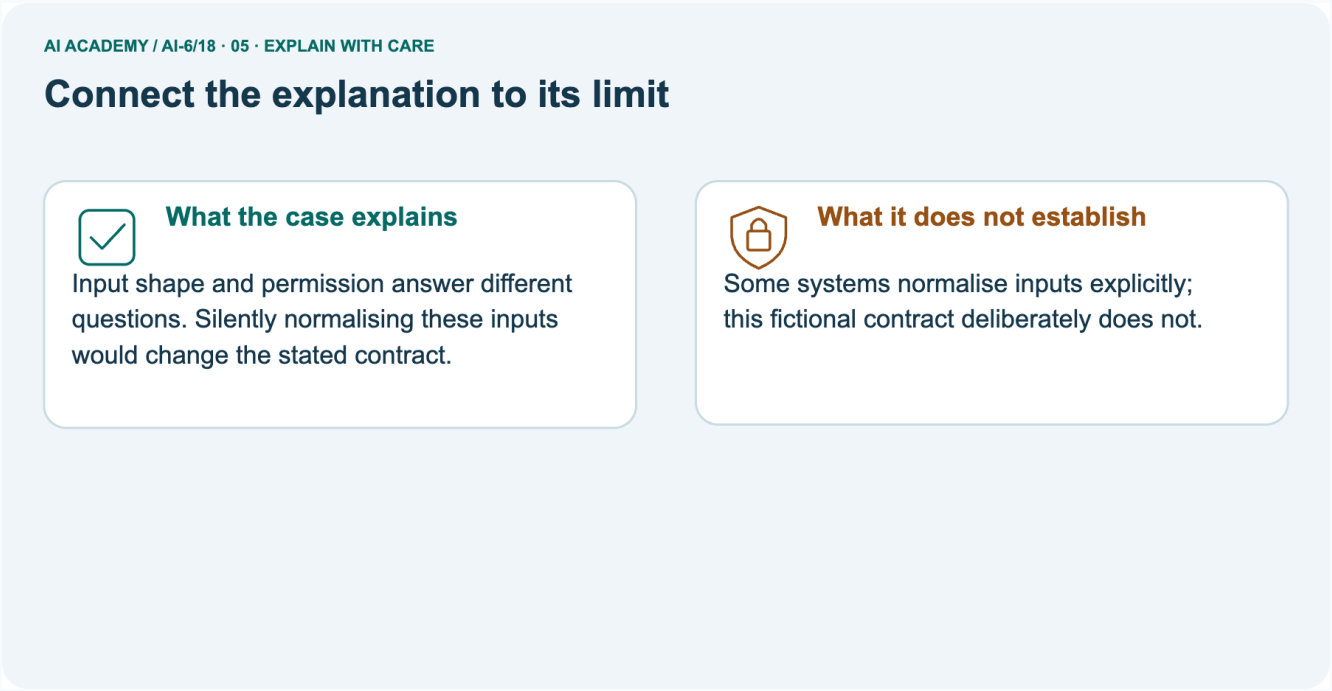
Reason: Structure contains no **evidence** of ownership or user permission.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Structure contains no **evidence** of ownership or user permission.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

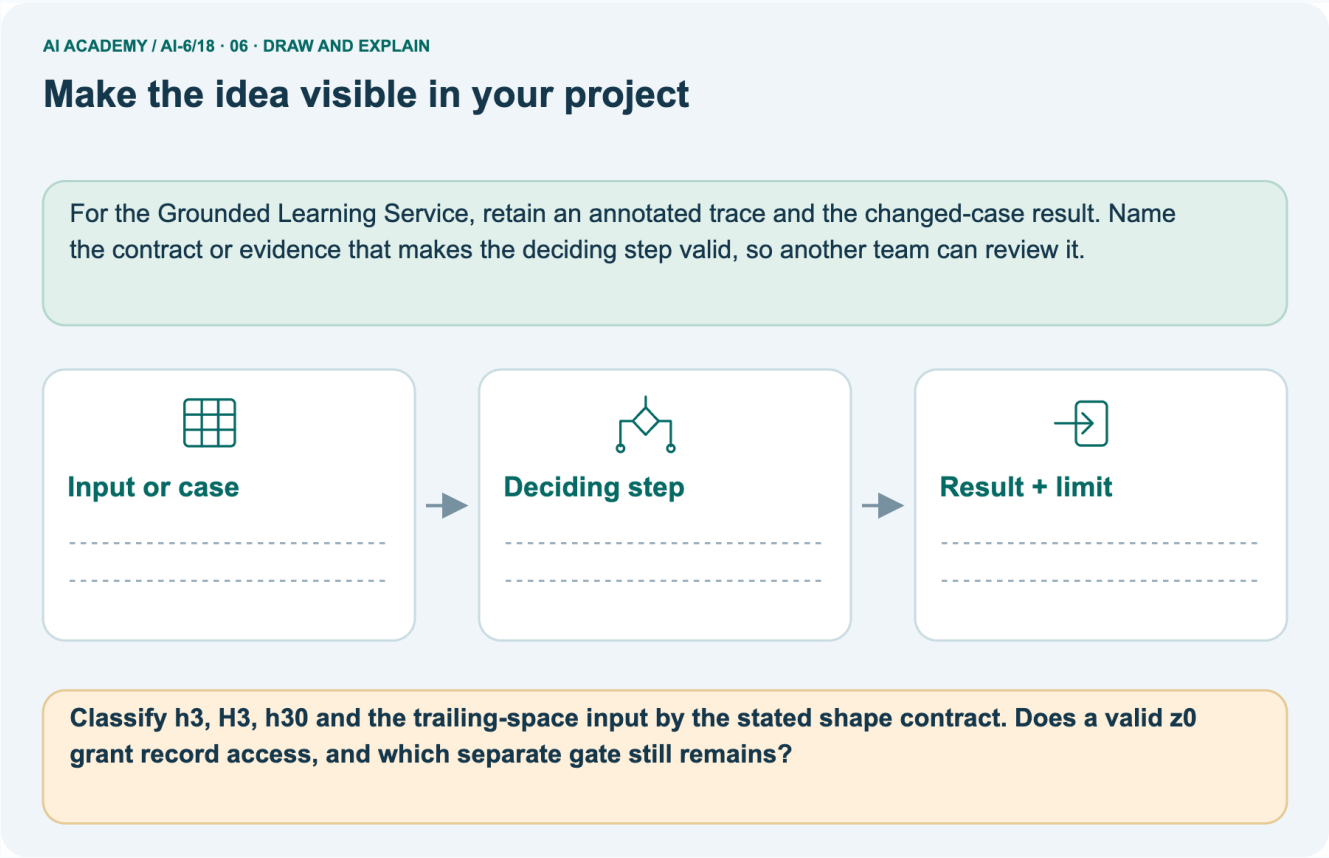
Input shape and permission answer different questions. Silently normalising these **input**s would change the stated contract. Limit: Some systems normalise **input**s explicitly; this fictional contract deliberately does not.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Some systems normalise **input**s explicitly; this fictional contract deliberately does not.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Classify h3, H3, h30 and the trailing-space **input** by the stated shape contract. Does a valid z0 grant record access, and which separate gate still remains?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Two characters with the required case and digit pass. H3 has wrong case; h30 and the trailing-space **input** have wrong length. An accepted identifier still needs trusted authorisation.

Changed case: The shape passes; resource access is still undecided. Structure contains no **evidence** of ownership or user permission.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Untrusted input boundaries: the case

02. Follow the deciding steps

03. Untrusted input boundaries — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.